Preliminary Sample-Size Analysis for CLAPS

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Abstract

CLAPS asks whether the acceptability of a passive sentence depends on how strongly the event affects the person or thing it happens to, and whether that pattern holds across languages. This report asks how large a sample is needed to answer that convincingly. The answer turns on an unusual feature of the design. Affectedness is a property of the verb, not of the participant, so precision is governed by how many verbs are tested rather than by how many people rate them. Past a point, recruiting more participants stops helping.

At the 72 verbs used here, English's theoretical infinite-participant ceiling is 81%, only just above the conventional 80% target, and its finite-sample simulations remain below target. Turkish's ceiling is itself below 80%. Norwegian is the exception, reaching target at about 100 participants. Earlier simulations were more optimistic for a correctable reason. They assumed less variation between verbs than the pilot data show, and omitted the variation in how individual verbs behave across sentence types. Measured rather than assumed, both push the required samples up sharply.

Two remedies fit within what the project can realistically collect, 80 participants per language at the current verb count. Analysing the three languages together roughly triples the verb-level information and lets every language contribute, Turkish included. Adopting a moderate rather than a strong standard of evidence, as the project proposed at the outset, brings English and Norwegian to target individually and makes the combined test near-certain. Turkish meets neither standard alone and is better reported as an estimate than as a test. Simulations continue, but cannot change this picture, since every per-language sample at or below the agreed cap is simulated to full replication and the cells still accumulating all lie above it.

This note reports preliminary sample-size guidance for CLAPS (A Crosslinguistic Approach to Passive Semantics). The guidance comes from a Bayes-factor design analysis. A design analysis of this kind simulates the planned model many times under plausible true effects, then asks how often a given sample size would yield compelling evidence for the focal predictions.

Two versions of the analysis are run. The primary one is data-grounded. It fits the planned model to each language's real pilot data and simulates new studies from that fit, so its results reflect the effects actually observed in the CLAPS pilot. The collaborators asked the design analysis to use that pilot as its basis. A single pilot's point estimate is noisy and tends to overstate power, the concern that Albers and Lakens (2018) raise for pilot-based power analyses. The data-grounded analysis avoids that point estimate in two ways. First, on each simulated study it draws the focal effects from the pilot posterior, the range of effect sizes the pilot data still leave plausible, so that the reported probability of a decisive Bayes factor reflects the pilot's uncertainty about the size of the effect. This yields an assurance, a power averaged over the plausible sizes of the effect, rather than a power computed at one assumed size. Second, it also evaluates a deliberately conservative lower bound on the effect, as a safeguard against an optimistic pilot. A second version, for comparison, is based instead on the pooled effect sizes from the five previously published studies. It provides an independent cross-check against the broader evidence base.

The two versions agree on the central structural point. The power to detect the affectedness effect is governed by the number of verbs in the materials rather than by the number of participants, because affectedness is a property of the verb. They diverge on whether a modest sample is enough. On the published-literature effects, a per-language sample of at most 50 participants would suffice at the full verb set. On the pilot effects, and with the verb-level variability measured from the pilot rather than assumed, the picture is less favourable. Norwegian reaches the target at about 100 participants, but English and Turkish do not reach it at any simulated sample size. For those two languages, the binding constraint is the number of verbs, and in Turkish also the strength of the effect itself, so adding participants alone cannot bring the design to the usual power target. Because the data-grounded analysis is the agreed primary basis, this is the finding that should guide the design. The report quantifies these limits analytically, explains why the two analyses disagree, and sets out design options that remain feasible for a volunteer consortium. The recommended option is a pooled cross-linguistic analysis, and the section on it below gives its rationale and the first results.

The data-grounded analysis is substantially complete. Results are in for all three languages up to 130 participants, for English up to 400 and for Turkish up to 400, and the largest samples are still computing. Each simulated condition rests on 22 to 144 studies, so the individual figures carry a Monte-Carlo error of several percentage points, that is, the uncertainty that comes from basing each figure on a limited number of simulated studies. The 400-participant simulations now confirm the English plateau, the levelling-off of power short of the target, so it no longer rests on extrapolation. The report should be read as a decision-relevant interim pass rather than a final recommendation.

Background and Motivation

The question CLAPS addresses is whether the acceptability of passive and related sentence types tracks the semantic affectedness of the patient, and whether that relationship differs by sentence type. Affectedness, the degree to which a patient or theme undergoes a change of state or is causally impinged upon by the action, is a core dimension of argument structure (Beavers, 2011; Dowty, 1991). It has long been argued to be the semantic heart of the passive, which carries the meaning that the surface subject is in a state or circumstance characterised by the agent having acted upon it (Pinker et al., 1987). Across languages, adult acceptability-judgement and comprehension studies support the view that passive acceptability is semantically structured rather than purely syntactic (Ambridge et al., 2016; Ambridge et al., 2023; Aryawibawa & Ambridge, 2018; Bidgood et al., 2020; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021).

The cross-linguistic evidence base to date spans English, Indonesian, Balinese, Hebrew and Mandarin. A Bayesian meta-analytic synthesis finds a consistent pattern across them (Ambridge et al., 2023). Affectedness raises acceptability for both passives and actives, more strongly for passives, and does not raise it for pseudo-passives. The affectedness constraint has not yet been tested in Turkish or Norwegian. CLAPS extends the paradigm to these two typologically and morphologically distinct languages. No prior language-specific effect-size estimates exist for them, so the CLAPS pilot supplies the first such estimates. The data-grounded analysis uses these pilot estimates as its primary basis, and the cross-linguistic literature provides an independent comparison, which turns out to be the more optimistic of the two.

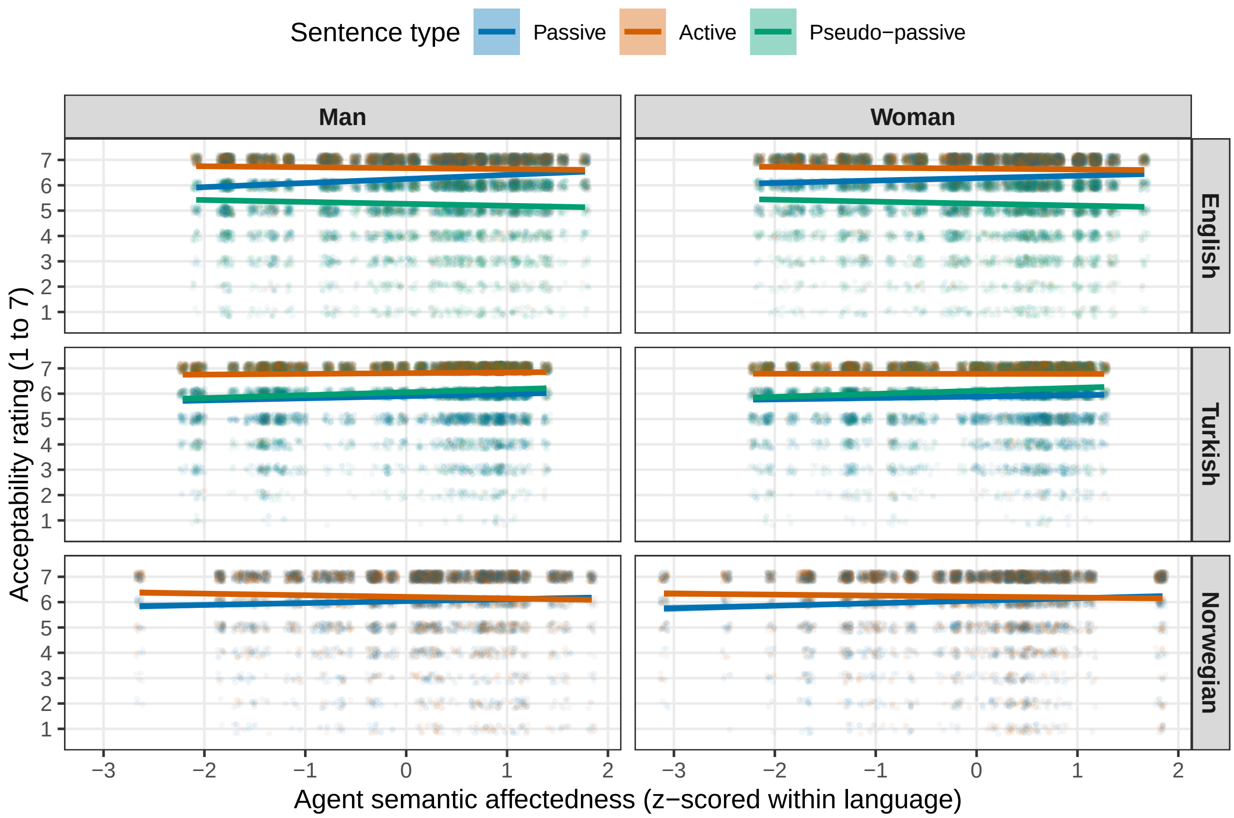
The Acceptability Model and the Focal Predictions

Each acceptability judgement is a discrete rating from 1 to 7. These ratings are modelled with a Bayesian cumulative-logit mixed-effects model, an ordinal model that respects the discreteness, ordering and bounded range of rating data while leaving the underlying scale free to vary across conditions (Bürkner & Vuorre, 2019). Treating the same ratings as continuous, under a Gaussian likelihood, biases the regression coefficients and inflates error rates where floor or ceiling effects are present (Liddell & Kruschke, 2018). Recent empirical work shows that modelling them cumulatively recovers psycholinguistic effects more accurately (Taylor et al., 2023; Veríssimo, 2021).

The expected pattern is already visible in the pilot data, which come from the sample whose composition by language is given in Table 4 in Appendix A. Figure 1 shows the raw acceptability ratings against agent semantic affectedness, separately by sentence type, referent gender and language. The affectedness slope is positive and steepest for the passive in English and Norwegian, and shallower for the active and the pseudo-passive. The trends are very similar for male and female referents within each language. This is reassuring, because the focal effects should not be driven by a referent-gender imbalance.

Figure 1

Raw Pilot Acceptability by Agent Semantic Affectedness, Sentence Type, Referent Gender and Language



Note. Points are individual pilot ratings, jittered. Lines are per-sentence-type linear trends with 95% confidence bands. The pilot data are used for calibration only. Norwegian has no pseudo-passive.

The model predicts acceptability from sentence type, from verb affectedness, and from their interaction. The passive is the reference level, and the active and the pseudo-passive are the comparisons. The focal predictions concern the affectedness slope in the passive and how it changes across sentence types. The primary prediction (H1b) is one-tailed and holds that affectedness raises passive acceptability more than it raises active acceptability, so the active-by-affectedness interaction is negative. A supporting prediction (H1a) holds that affectedness raises acceptability in the passive itself, so the passive affectedness slope is positive. A sample size is judged adequate only when both of these predictions reach the evidence threshold. A secondary, two-tailed prediction (H2) concerns the pseudo-passive-by-affectedness interaction. It does not gate the recommendation and does not apply to Norwegian, which has no pseudo-passive.

Design-Analysis Method

The design analysis estimates statistical power by simulation. Each simulated dataset is drawn from the same cumulative-logit model that will be used for analysis, under assumed values for the effects of interest. Each dataset is then analysed in the same way. The proportion of simulated studies that yield decisive evidence estimates the power at that sample size. The whole procedure stays within the Bayesian framework. This matters because the confirmatory test is a Bayes factor rather than a significance test, and the power to return a decisive Bayes factor is not the same quantity as the power to return a significant p-value.

A central feature of the design is that affectedness is a property of the verb rather than of the individual trial. Each verb carries a single affectedness value, held constant across participants and sentence types, exactly as in the real materials. The number of verbs, rather than the number of trials, therefore determines the precision with which the affectedness effect can be estimated. The verb count is treated as a design dimension alongside the participant count.

The two analyses differ in where they take the assumed true effects and the affectedness values. The data-grounded analysis, which is primary, fits the planned model to each language's pilot data. It then takes the population effects, the random-effect covariance and the ordinal thresholds from that fit, and it reads each verb's actual affectedness score from the pilot data. The literature-anchored cross-check instead sets the population effects to values from the cross-linguistic meta-analysis. On the standardised affectedness predictor, the passive affectedness slope is set near the pooled value of about 0.23 logits per standard deviation (a logit is a unit of the model's underlying scale, not a raw point on the 1 to 7 rating), with a negative active-by-affectedness interaction of about -0.15 and a small positive pseudo-passive-by-affectedness interaction. In the cross-check, affectedness is spread across a plausible range rather than read from the pilot, and the by-participant and by-verb random-effect standard deviations are set to realistic magnitudes. The referent-gender covariate is held out of both preliminary runs and is examined separately.

The analysis model uses the maximal random-effects structure consistent with the design, meaning the most complex structure the design supports, so any effect that can vary across participants or verbs is allowed to do so. It has correlated by-participant slopes for sentence type, affectedness and their interaction, together with by-verb intercepts and sentence-type slopes (Barr et al., 2013). Because affectedness does not vary within a verb, a by-verb affectedness slope cannot be estimated and is omitted, so affectedness enters as a population-level effect, one estimated for the sample as a whole rather than allowed to vary by verb. The priors are weakly to moderately informative and centred at zero on the focal slopes, so that the directional tests are not pre-loaded. A normal prior of standard deviation 0.5 is placed on the affectedness slope and on the active interaction, and a slightly wider normal prior of 0.6 on the pseudo-passive interaction. The sentence-type main effects take a wider normal prior. Student-t priors are placed on the ordinal thresholds and on the by-participant and by-verb standard deviations, and an LKJ prior, the standard prior for a correlation matrix, on the correlations among those random effects (Gelman, 2006; Lewandowski et al., 2009).

Evidence for each prediction is summarised by a directional Savage-Dickey Bayes factor, a number that weighs how strongly the data favour the predicted direction of the effect over there being no effect, obtained by sampling the prior alongside the posterior (Wagenmakers et al., 2010). A simulated study counts as a detection when its Bayes factor reaches 10, the preregistered threshold for strong evidence. Under the zero-centred prior, this is not a stringent bar. The Bayes factor used here is directional, meaning the posterior odds that the coefficient carries the predicted sign divided by the same odds under the prior, so under a symmetric prior a value of 10 is odds of 10 to 1 on the direction. That is a posterior probability of 0.91, or about 1.34 standard errors from zero, whereas a one-sided $p < .05$ requires 1.64, so the Bayesian criterion is if anything the less demanding of the two. It is not the point-null Bayes factor of a default two-sided test, which is a different quantity on a different scale. Power is the proportion of simulated studies that reach it. The recommended sample size is the smallest at which both focal predictions reach a power of 80%.

Each replicate is fitted with four Markov chains of 3,000 iterations after 1,000 warm-up iterations, at the same target acceptance rate of 0.99 and maximum tree depth of 12 that the confirmatory analysis will use. The proposal specifies a longer run for that single confirmatory fit, at sixteen chains of 5,000 iterations after 2,000 warm-up iterations. The shorter chains add Monte-Carlo noise to each simulated Bayes factor rather than shifting it systematically, so the power figures below are, if anything, mildly conservative.

The literature-anchored grid crosses the three languages with participant counts from 50 to 120. The output read here covers English, Norwegian and Turkish from 50 to 120 participants at verb counts of 40 and 72. Each completed cell holds 50 replicates, which gives a Monte-Carlo standard error of about five to six percentage points near a power of 80%, so estimates close to the threshold are not perfectly ordered across adjacent sample sizes. The central code for the data-generating process, the model and the Bayes factor is reproduced in Appendix B. The full operating characteristics of this run, cell by cell, are reported in Table 5 in Appendix C, and Appendix D maps the whole repository, including which file implements each of the two simulation engines described above and how the pooled analysis reuses one of them.

Results

Data-grounded results

The data-grounded analysis is the primary basis for the recommendation, and it is now substantially complete. In its assurance form it draws the focal effects afresh from each language's pilot posterior on every simulated study, so the reported power carries the pilot's own uncertainty about the size of those effects. Table 1 reports the joint power, the proportion of simulated studies in which both focal predictions clear the Bayes-factor threshold together, at the pilot's real verb count.

Table 1

Preliminary Data-Grounded Joint Power by Participant Count and Language, Assurance Variant

Participants	English	Turkish	Norwegian
30	25.0%	20.8%	41.7%
50	41.7%	37.5%	66.7%
70	47.9%	36.8%	68.8%
80	60.0%	41.7%	68.3%
100	55.6%	38.2%	81.9%
130	45.8%	29.2%	87.5%
150	62.5%	35.0%	-
200	65.0%	42.5%	-
250	70.0%	42.5%	-
300	72.5%	35.9%	-
400	62.5%	40.9%	-

Note. Joint power is the proportion of studies in which both focal predictions clear a Bayes factor of 10 together, at 72 verbs and 66 for Norwegian. Cells below 10 replicates are still accumulating and shown blank. The rest hold 22 to 144 studies each. Percentages are shown to one decimal place to preserve the exact Monte-Carlo summaries. Norwegian at 130 participants sits a little above the approximate infinite-participant ceiling in Table 2, at $21/24 = 87.5\%$ against a Monte-Carlo standard error of 6.8%, which is sampling variation rather than a disagreement between the two.

The pattern differs sharply by language. Norwegian reaches the 80% target at about 100 participants and is comfortably powered beyond it. English and Turkish do not reach it at any simulated sample size. For English, the analytic ceiling is 81%, but that is the limiting value with unlimited participants, not the power of a finite study; its simulated joint power remains below 80%. It is 72.5% by 300 participants and 62.5% at 400. The reason is structural. The joint requirement is held back by the affectedness main effect (H1a), whose detection rate levels off at roughly three in four, while the interaction (H1b) climbs comfortably past it. Because affectedness is a property of the verb, its main effect is estimated from the 72 verbs, so adding participants does not sharpen it. The precision of that effect, and with it the power to detect it, is set by the number of verbs. For Turkish, both focal effects are weak in the pilot. The joint power never exceeds 42.5% at any simulated sample size, and the present design does not bring it to target.

A conservative safeguard variant, which fixes each focal effect at a cautious lower bound of its pilot posterior, is lower again. Its highest joint power over the samples run is 58% for Norwegian, 25% for English and 12% for Turkish. This is the design's power under a deliberately pessimistic reading of the pilot.

The pilot posterior itself puts the probability that both focal effects have their predicted sign at 99.6% for English, 99.5% for Norwegian and 94.0% for Turkish, so the pilot points the right way in all three languages. The shortfall for English and Turkish is therefore a matter of precision rather than direction. Because affectedness varies only between verbs, the standard error of its effect cannot fall below the by-verb variability divided by the spread of the affectedness scores, however many participants respond. This is the stimulus-sampling limit long recognised in psycholinguistics (Clark, 1973) and quantified for designs with crossed random effects by Westfall et al. (2014; see also Judd et al., 2017; Brysbaert & Stevens, 2018). The limit can be computed directly from the fitted pilot models, with no simulation involved. Table 2 reports the resulting upper bounds, the detection rates that would remain even with unlimited participants at the current verb counts.

Table 2

Analytic Upper Bounds on Detection with Unlimited Participants, at the Current Verb Counts

Language	H1a ceiling	H1b ceiling	Joint ceiling	Verbs for 80%	Verbs for 90%
English	83%	97%	81%	72	124
Turkish	77%	65%	49%	348	$> 1{,}200$
Norwegian	85%	100%	85%	66	98

Note. Ceilings are the probability of a Bayes factor of at least 10 with unlimited participants, from the 8,000 pilot posterior draws and the fitted by-verb covariance at 72 verbs and 66 for Norwegian. The last columns give the verbs needed for an 80% and 90% joint ceiling. A ceiling just above target, as for English, still leaves it out of reach, since any finite sample sits below its ceiling.

These bounds agree closely with the simulations. English's simulated joint power climbs to 72.5% by 300 participants against a ceiling of 81%. Turkish never exceeds 42.5% against a ceiling of 49%. Norwegian's simulated 81.9% at 100 participants matches its ceiling of 85% within Monte-Carlo error. The agreement between two entirely different routes to the same quantities, stochastic simulation and direct calculation, is a strong check that the pipeline is sound. A further check makes the point starkly for English. The pilot's own posterior standard deviation for the affectedness slope, which measures how precisely the pilot pins that slope down, is 0.19, already close to the verb-count floor of 0.19. A replication with the same 72 verbs would therefore have little room to estimate that slope more precisely than the pilot already has, whatever its sample size. The residual gap reflects the floor formula's own approximation, since it ignores some of the covariance in the by-verb random effects, rather than a real target that more participants could close.

The ceilings also show what would and would not help. Relaxing the evidence criterion from a Bayes factor of 10 to 6 or 3 lifts the English joint ceiling to 86% and 92%, but lifts Turkish only to 58% and 70%, so a weaker threshold cannot rescue the Turkish test. Adding verbs raises every ceiling, because the precision floor shrinks with the square root of the verb count. English would reach a joint ceiling of 90% at about 124 verbs, and Norwegian at about 98. Turkish would need about 348 verbs merely to reach a ceiling of 80%, which is not realistic within a single language, so for Turkish the remedy lies elsewhere, as taken up under Paths Forward below.

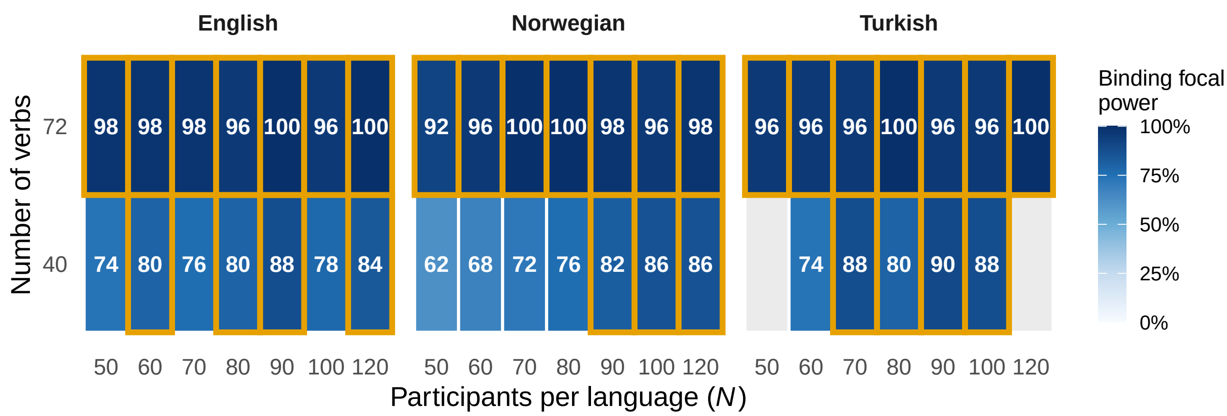
These results are preliminary in one remaining respect, since each condition rests on 22 to 144 simulated studies and so carries a Monte-Carlo error of a few percentage points at the better-populated sample sizes and more at the sparser ones. The English plateau itself, though, is now confirmed rather than extrapolated. At 400 participants, with 32 replicates in, the joint power sits at 62.5% (20/32), compared with 72.5% (29/40) at 300 participants. These sparse estimates are not literally flat, but their decline is compatible with Monte-Carlo variation and, critically, there is no sustained rise towards the target. Overturning the plateau would have required a sustained rise at 400 and 500, and the 400-participant data already rule that out. The largest cells still computing will fill in the remaining sample sizes, but by that test they can no longer overturn the plateau.

Literature-anchored cross-check

The grid is read as a surface over participant count and verb count, summarised by the binding focal power, the smaller of the two focal detection probabilities, since the recommendation requires both to clear 80%. This cross-check is the more optimistic of the two analyses, because it fixes the effects at the pooled literature values and carries no uncertainty about them. Figure 2 shows this surface for the two languages whose cells are complete. Cells outlined in gold meet the target, and blank cells are those whose replicates are still accumulating. The pattern is clear. Power increases steeply with the number of verbs and only weakly with the number of participants.

Figure 2

Binding Focal Power by Participant Count and Verb Count, for Each Language

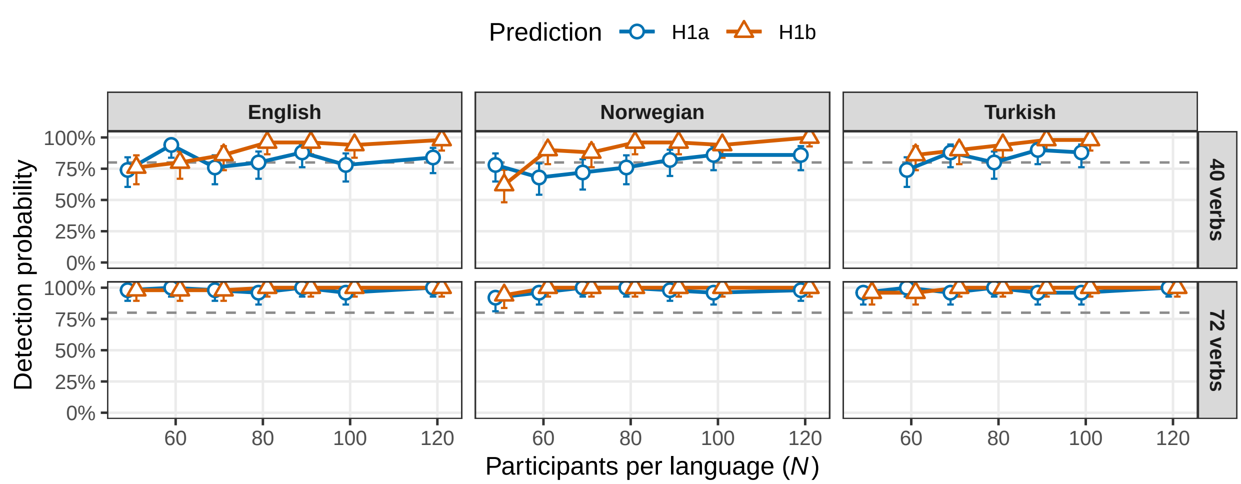


Note. Each cell gives the binding focal power as a percentage, with darker shading for higher power. Cells outlined in gold meet the 80% target. Blank cells are still accumulating.

The two axes are not equally important. At 40 verbs in English, the binding power is 74% at 50 participants and reaches the 80% target only between about 60 and 90 participants, within Monte-Carlo noise. At 72 verbs, it is already 92% to 98% at 50 participants. Increasing the number of verbs from 40 to 72 raises power more than increasing the number of participants does, which follows from affectedness being a verb-level property. Figure 3 separates the two focal predictions and shows how each approaches the target as the sample grows, at each verb count.

Figure 3

Detection Probability for the Focal Predictions, with Wilson 95% Intervals, by Verb Count and Language



Note. Points are detection probabilities over the simulated studies in each complete cell, with Wilson 95% intervals reflecting Monte-Carlo uncertainty. The dashed line marks the 80% target. Points are dodged so the two predictions do not overlap.

In the literature-anchored output, which covers English, Norwegian and Turkish, the binding focal power at the full 72-verb set is already between 92% and 98% at 50 participants, the smallest sample simulated, in every language included in this cross-check. On the literature effects, the adequate per-language sample is therefore at most 50 participants at the full verb set. This is the optimistic reading. The data-grounded results above, which condition on the weaker pilot effects, do not reach the target for English or Turkish at any sample size. Because 50 is the smallest sample in the main grid, the analysis places the adequate sample at or below 50 without determining it more precisely. Pinning it down further, below 50 participants, is not pursued here, since the recommendation rests on the data-grounded analysis rather than on this cross-check.

Why the Two Analyses Disagree

The gap between the two analyses calls for an explanation, because the literature-anchored grid reports 92% to 98% at 50 participants on the same 72 verbs at which the data-grounded joint power never reaches 80%. Expressed per standard deviation of affectedness, the pilot's passive slope is 0.26 logits for English, 0.23 for Norwegian and 0.18 for Turkish, against a pooled literature value of about 0.23. The Turkish active interaction, at −0.15, matches the literature's −0.15 almost exactly, and the English interaction, at −0.44, is almost three times the literature value, so the CLAPS pilot effects are not unusually small.

The disagreement comes instead from the noise side of the simulation, in two specific assumptions. First, the literature-anchored simulator assumed a by-verb intercept standard deviation of 0.5, a measure of how much verbs differ in their baseline acceptability, where the fitted pilot models put it at 0.83 for English, 0.64 for Turkish and 0.66 for Norwegian. The precision floor on the affectedness slope scales directly with this quantity, so assuming 0.5 where the fitted value is 0.83 overstates the attainable evidence considerably. Second, the literature-anchored simulator included no by-verb sentence-type slopes at all, allowing no verb-to-verb variation in how sentence type shifts the ratings, where the fitted pilots put their standard deviation at 0.88 for English, 0.64 for Turkish and 0.34 for Norwegian. That omission left the interaction test with no verb-level noise floor in the simulation, which is why the earlier grid found the interaction so easy to detect. Both assumptions were reasonable defaults before the pilot models existed. The pilot has now measured the quantities they stood in for, and the data-grounded analysis uses the measured values. The earlier optimism, in short, came less from the assumed effect sizes than from understating how much verbs vary (Clark, 1973; Westfall et al., 2014).

Effect-Size Qualification

One assumption deserves particular care, because it bears directly on the absolute sample size. The numbers shown here condition on assumed effects set to the cross-linguistic pooled posterior means from a Bayesian meta-analytic synthesis of the passive-affectedness literature, which is a central estimate rather than an optimistic one (Ambridge et al., 2023). On the simulator's internal predictor scale, these appear as larger nominal values, but expressed per standard deviation of affectedness, they are about 0.23 and −0.15 logits, matching the pooled literature values. The larger nominal numbers therefore reflect the scaling rather than a larger effect. Two cautions nonetheless apply. The first is the one the previous section sets out. The fitted pilot models put the verb-level variability well above the values this grid assumed, and that difference, more than any difference in effect size, is what made the grid optimistic. The pilot effects themselves are broadly in line with the pooled literature values once both are expressed per standard deviation of affectedness, though the Turkish terms sit at the weak end, with the interaction at about −0.15 in those units. The second is that the literature is itself subject to publication bias and to the winner's curse, the tendency of published estimates, especially from small and low-powered studies, to overstate true magnitudes (Button et al., 2013; Gelman & Carlin, 2014; Ioannidis, 2008). Design analyses anchored to such estimates can therefore be optimistic about power (Albers & Lakens, 2018; Vasishth et al., 2018).

These cautions are the reason the data-grounded analysis, rather than this cross-check, is the primary basis for the recommendation. It draws the focal effects from each language's own pilot posterior, and it also evaluates a conservative lower bound. It therefore conditions on the effects observed in CLAPS and carries their uncertainty through to the reported power. Its results are reported above. They agree with this cross-check for Norwegian, which is powered at about 100 participants, and they diverge from it for English and Turkish, where the weaker pilot effects hold the joint power below target at every sample size run.

A separate point concerns the scale on which the prior sits, rather than the size of the assumed effect. The literature-anchored grid behind the figures expresses affectedness on the simulator's internal predictor, whose standard deviation is about 0.29. The real confirmatory analysis and the data-grounded analysis both place it on the analysis pipeline's Gelman scaling, which divides a predictor by twice its standard deviation and so leaves it with a standard deviation of 0.5, and the same weakly informative prior is applied throughout. A Bayes factor reflects the width of the prior relative to the coefficient it constrains. Holding the prior fixed while the predictor scale differs therefore leaves the literature-anchored grid mildly optimistic about the focal-slope Bayes factor, by a factor of about 1.7, relative to the analysis that will actually be run. The qualitative conclusion is unaffected, because the verb count dominates the result, the focal Bayes factors at the full verb set lie far above the threshold, and the data-grounded analysis already uses the correct scale. A scale-corrected version of the cross-check grid is being computed to confirm the per-language sample sizes directly. Until it completes, the figures here should be read as a mild upper bound on the Bayes-factor evidence.

For an effect that lives at the level of the participant, a weaker true effect simply needs a larger sample, and the rough guide is that a true effect around two-thirds of the assumed magnitude roughly doubles the sample needed. The affectedness main effect lives at the level of the verb, however, where its precision is set by the number of verbs, so a weaker true effect cannot be recovered by recruiting more participants. This is exactly what the data-grounded results show. At the full verb count and the pilot's own effects, the affectedness power for English settles below the target and does not climb with sample size, and for Turkish neither focal effect is strong enough to bring the joint power to target. The effective levers are therefore the number of verbs, and for Turkish the strength of the interaction, not the participant count. Where feasible, the design should still be planned against a deliberately conservative, or safeguard, effect size rather than a point estimate (Perugini et al., 2014), and the data-grounded analysis builds this in directly through its assurance and safeguard variants. Discounting the literature-anchored grid towards weaker effects makes the same point: re-evaluated at 75% and 60% of the assumed effect magnitude, with the 60% level approximating the weaker effects seen in the pilot, it needs larger samples. The data-grounded analysis conditions on those pilot effects directly, so the primary results above already reflect that discounting.

Convergence and Computation

Convergence is sound across the grid, with no divergent transitions anywhere, which are the warning sign of an unreliable fit. At the smaller verb count, essentially every fit meets the strict criteria, namely an R-hat below 1.01, which checks that the separate chains have settled on the same answer, together with bulk and tail effective sample sizes of at least 400, which measure how much independent information the draws carry. At 72 verbs, the maximal model occasionally falls just short of these criteria, in about one fit in ten at the smaller samples and up to one in five at the largest, where N is 80 to 90. It falls short on the R-hat or effective-sample-size thresholds rather than on divergences, so it reflects how efficiently the sampler is exploring rather than a fault in the model's specification. Restricting the power calculation to the fully converged fits leaves the estimates essentially unchanged. For English at 72 verbs and 50 participants, the binding focal power is 98% over all 50 fits and 98% over the 46 converged fits, so the recommendation does not depend on the difference.

Secondary Prediction

The pseudo-passive-by-affectedness interaction reaches a detection probability of about 58% to 88% at 72 verbs. This remains below the focal predictions and reflects its smaller assumed magnitude, and the opposite-direction test is near zero throughout, as expected. This prediction is secondary and two-tailed, and it does not constrain the sample-size recommendation. Full operating characteristics for all four predictions, including the convergence rates, are reported in Appendix C.

Paths Forward

The analytic bounds turn the shortfall into a set of concrete design choices. Four options follow, in roughly increasing order of departure from the current plan. Two of them have since been closed by the project, and are kept here because the reasoning that ruled them out is part of the record.

The constraints are now these. The design is capped at 80 participants per language, with 100 the absolute maximum any team would be asked for. The verb pool cannot grow beyond 72: the limit is the supply of suitable verbs in each language, not the length of a session, so the third option below is unavailable rather than merely expensive. Collection must be at a single sample size fixed and announced in advance, because the teams have to take a stated number to their ethics committees, which rules out the fourth option. Many of the teams have not run a study of this type before, and a design they cannot explain is not a design they can run.

The first option changes nothing about the materials and revisits the adequacy rule. The design's primary prediction is H1b, the active-by-affectedness interaction, with H1a as a supporting prediction. Gating each language on the primary prediction alone, with the passive slope reported as estimation, brings English to 80% on the interaction at 80 participants and Norwegian at 70, at the current verb counts and at the strong threshold. Those crossing points are the smallest samples at which the *lower bound* of the simulated rate reaches 80%, not the first cell whose point estimate happens to. With a few dozen replicates per cell the two can differ by a whole sample size. The joint gate on both predictions is what no feasible English sample can satisfy. This option does not help Turkish, whose interaction ceiling is 65% even on its own.

The second option makes the cross-linguistic claim the primary confirmatory test. The theoretical claim CLAPS addresses is explicitly cross-linguistic, that affectedness structures passive acceptability across languages, and the meta-analytic synthesis of the earlier studies frames the evidence in exactly those terms (Ambridge et al., 2023). A hierarchical model fitted to all three languages jointly tests that claim directly. It also multiplies the verb-level information, roughly 210 verb-by-language units in place of 72, which divides the precision floor by about the square root of three. On the pilot effects, every language's focal terms point the predicted way, so the pooled test starts from a favourable position. Turkish then contributes fully to the confirmatory result, and its language-specific effects are reported as estimates with credible intervals, which reflects honestly what a 72-verb single-language design can support. The pooled design analysis runs on the same machinery used here, and it is now built. Its rationale and the first results are set out below.

The third option expands the verb pool without lengthening any participant's session. Verbs, not participants, are the scarce resource, and a planned-missingness design assigns each participant a random subset of a larger pool (Graham et al., 2006; Little & Rhemtulla, 2013). Each volunteer still rates about 72 items, so the 45-minute session is unchanged, while the pool grows to, say, 120 to 150 verbs. At about 124 verbs, the English joint ceiling reaches 90%, and the pooled cross-linguistic test gains in proportion. The cost falls on materials, namely affectedness norms and translations for the new verbs, rather than on participant time. This option is closed. It would have required the 72-verb limit to be a constraint on the session, which planned missingness respects, rather than on the pool. It is the pool: the teams are already struggling to find 72 suitable verbs in most languages, so there is no larger set to sample from. That matters beyond this one option, because the verb count is what every ceiling in this report is a ceiling on. With the pool fixed, those ceilings are now permanent features of the design rather than quantities a bigger materials budget could move, and Turkish, which would need about 348 verbs to reach 80% on its own, is out of reach for good.

The fourth option spends volunteer capacity more efficiently rather than demanding more of it. A sequential design analyses the data in waves and stops as soon as the Bayes factor reaches a preset bound or the sample reaches an agreed maximum. This typically needs markedly fewer participants than a fixed-sample design with the same operating characteristics, meaning the same chances of reaching a decisive result (Schönbrodt et al., 2017; Stefan et al., 2019). Optional stopping of this kind poses no problem for Bayes-factor inference (Rouder, 2014).

This option is also closed, and for a reason worth recording, since the statistical objection to it is the one that does not apply. Nothing in the Bayes-factor framework penalises optional stopping. The obstacle is operational: the collecting teams have to put a specific number of participants in front of an ethics committee, and a protocol whose sample size depends on interim results is one that many of them could not explain or defend, however sound it is. A design that cannot be administered by the people running it is not available, whatever its operating characteristics. The sample size below is therefore a single fixed number, announced in advance.

The recommendation is the first two options taken together, and nothing else: the pooled cross-linguistic analysis is the confirmatory test, it is gated on the interaction alone, its threshold is a Bayes factor of 6, and every team collects 80 participants.

Gating on the interaction alone is what makes the design work, and the size of the gain is worth being precise about. It is not that the effects were too small. Requiring both focal predictions in the same study caps the result at the product of their two ceilings, and it is the passive-only slope that binds: it tops out at 83% in English and 85% in Norwegian however many participants are recruited, against 97.1% and 99.7% for the interaction. Dropping the conjunction lifts the English ceiling from 81% to 97.1% and the Norwegian ceiling from 85% to 99.7%. The conjunction was never doing theoretical work in any case. The claim CLAPS tests is that affectedness matters *more* for passives than for actives, which is the interaction; the passive-only slope is a check that the interaction means what it is supposed to mean, and it belongs in the report as an estimate rather than in the gate as a second hurdle. It is reported in full below, with its own detection rate, as a prespecified positive control.

The threshold needs a plainer justification than its name gives it, because the Bayes factor used here is directional and does not mean what the conventional verbal labels mean. A directional Bayes factor of k is reached exactly when the posterior probability of the predicted sign reaches $k/(k+1)$, so a Bayes factor of 3 is a posterior sign probability of 0.75, which is 0.67 standard errors from zero and corresponds to a one-sided tail probability of 0.25. A criterion that lenient will not carry a confirmatory claim through review, whatever it is called, and an earlier draft of this report recommended it. A Bayes factor of 6 is 0.14 on the same scale and 10 is 0.091; matching a one-sided 5% test would need about 19. The threshold is set at 6 because it is the strongest criterion the design can meet at the cap while leaving the confirmatory test adequately powered, and because stating its tail equivalent alongside it is more honest than borrowing a label from a different quantity.

At that threshold the pooled test detects the interaction in 84% [0.76, 0.90] of simulated studies, combining the pooled cells at or below the cap. Cell by cell the pooled arm carries only 16 to 30 replicates and no single cell settles anything; combined, and given a design in which power is flat in sample size, that is the best estimate available. At a Bayes factor of 10 the same figure is 57% [0.47, 0.67], which is why 10 is not the recommendation.

Each language is then reported against the same criterion, as an estimate with its full posterior rather than as a second gate. English reaches 96% [0.91, 0.99] on the interaction at 80 participants and Norwegian 98% [0.93, 0.99], so a team publishing a single-language study in either can state a conventional power figure for the confirmatory prediction and defend it. Turkish reaches 64% [0.55, 0.73], and its ceiling on the interaction is 65% at a Bayes factor of 10 however many participants are recruited. Turkish contributes fully to the pooled confirmatory result and is reported as estimation on its own. That is not a shortfall to be fixed by recruitment; it is what 72 verbs buy in a language whose interaction is the weakest of the three. Turkish is also the one language whose passive-only slope is the better-detected of its two focal predictions, 75% against 57% at a Bayes factor of 10. Anyone reporting Turkish on its own should lead on the slope rather than the interaction.

Eighty participants is the recommendation, and 100 would not change it. Between those two samples the simulations differ by less than their own resolution, and the analytic bounds say the most that 20 further participants could add is a few points, because the binding information is in the verbs. The case for 80 is that it is the number the teams can collect.

What Is Settled and What Is Still Open

The design analysis varies several things at once, and it is worth separating the questions the results have answered from those that remain a matter of judgement. Three findings are settled and no further simulation will overturn them. Power for the affectedness effect is governed by the verb count rather than the participant count, so recruitment alone cannot rescue a language that is short of target. English and Turkish do not reach the conventional target on both focal predictions at any feasible per-language sample, whereas Norwegian does. Pooling the three languages multiplies the verb-level information the estimate rests on and is the only remedy that stays inside the agreed cap without changing the materials.

The choices that follow from this are set out in Table 3. Each row is a decision for the collaborators, with what the evidence says about it and what, if anything, is still outstanding.

Table 3

Design decisions, with what settled each.

Choice	Decision	What settled it
Adequacy rule	The interaction alone	The conjunction of both focal predictions is what no feasible sample reaches; dropping it lifts the English ceiling from 81% to 97%. The passive-only slope is reported as a positive control
Evidence threshold	Bayes factor of 6	The strongest criterion the design meets at the cap. A directional Bayes factor of 3 is a one-sided tail probability of 0.25, too lenient to carry a confirmatory claim; 10 leaves the pooled test at 57%
Sample size	80 per language	Between 80 and 100 the simulations differ by less than their own resolution, and the verb count binds. 80 is what the teams can collect
Confirmatory test	Pooled cross-linguistic	The theoretical claim is cross-linguistic, and the verb-level information is greatest there
Turkish	Estimation only	Its ceiling on the interaction is 65% at a Bayes factor of 10 whatever the sample. Settled; further cells will not change it
Verb pool	72 verbs, fixed	Closed by the project: the limit is the supply of suitable verbs, not session length, so the pool cannot be enlarged
Stopping rule	Fixed sample, announced in advance	Closed by the project: teams must take a stated sample size to their ethics committees. The statistical objection to sequential testing does not apply, but the operational one does

None of these now waits on further computation. Two were closed by the project rather than by the simulations, and are recorded as such. The rest follow from results already in hand, and the simulations have said everything they can about them.

One caveat belongs with the whole table. Every rate in this report is computed over all fitted cells, whether or not they pass the convergence screen, which is the honest choice because a real analysis would not discard a study for imperfect sampling. The proportion passing that screen falls as the sample grows, so rates at different sample sizes are averages over somewhat different mixtures of clean and untidy fits. On the screened subset the ordering of adjacent sample sizes sometimes reverses. This does not affect the recommendation, which rests on a criterion the design clears with room to spare and on a sample size chosen for feasibility, but it is a reason to read small differences between adjacent cells as noise rather than as structure.

The Pooled Analysis: Rationale and Expectations

Following the recommendation above, the pooled cross-linguistic design analysis has been implemented and validated. Each simulated study draws every language's effects from that language's own pilot posterior, exactly as in the per-language assurance runs, and simulates the three datasets. It then analyses them together with the cross-language model used in the earlier synthesis work, which carries correlated random slopes by participant, by verb and by language. The evidence criteria are unchanged, namely directional Savage-Dickey Bayes factors on the pooled focal coefficients under the same zero-centred priors. Under the 80-participant-per-language cap set by the project lead, the pooled grid covers samples from 50 to 100, and all simulation re-runs are capped at 100 participants and 72 verbs, with the larger runs cancelled. The decision arm, at 70, 80 and 100 participants, carries 120 per cell, which brings the Monte-Carlo error near a power of 80% down to about four percentage points. The pooled analytic ceiling given below is computed from the fitted per-language models rather than from a single pooled fit. A fit of the full pooled model to the combined real pilot data, as a check that the model settles on real rather than simulated data, remains outstanding. The first pooled cells have returned, and they separate the two evidence thresholds sharply. At 50 participants per language, across the 30 replicates available so far, the joint power is 37% at the strong-evidence threshold but 97% at a Bayes factor of 3, with median Bayes factors of 13 and 11 for the affectedness slope and the interaction. Pooling therefore makes moderate evidence for both predictions close to certain within the cap, while strong evidence for both together falls well short of the conventional target. Restricting the estimate to fits that pass every convergence check moves it by a few percentage points in either direction and does not change that conclusion. These cells carry a Monte-Carlo error of about 9%, and the larger samples are still computing, so the figure will firm up but is unlikely to reverse.

The rationale is theoretical before it is statistical. The hypothesis CLAPS tests is cross-linguistic, and the evidence for it has always been read across languages, most explicitly in the meta-analytic synthesis of the five earlier studies (Ambridge et al., 2023). The per-language analyses treat each language as a self-contained confirmatory study, which is exactly why each inherits its own verb-count ceiling. The pooled model instead tests the general claim directly. Its population-level affectedness slope and interaction are the cross-linguistic effects, and its by-language random slopes let each language depart from them, so generality is tested rather than assumed.

Statistically, pooling eases the binding constraint identified above. The precision floor on the affectedness effect is set by the verb-level information, and the pooled analysis draws on about 210 verb-by-language units instead of 72, which shrinks the floor by roughly the square root of three. The pilot slopes point the predicted way in all three languages, at 0.26, 0.18 and 0.23 logits per one within-language standard deviation of affectedness for the main effect and −0.44, −0.15 and −0.27 on the same scale for the interaction, so their average, which leans on the more precisely estimated languages, sits comfortably clear of zero. Partial pooling also shares strength in the other direction, since each language's estimate borrows from the others in proportion to its own uncertainty, and Turkish gains most from that. The fitted coefficients on the model's Gelman-scaled predictor are about twice these per-standard-deviation values because that predictor has a standard deviation of approximately 0.5.

The expected differences from the per-language results follow from this. First, the ceilings should effectively disappear as constraints. With the pooled effects near the averages above, and a precision floor of about 0.05 logits per standard deviation, both pooled focal ceilings sit near 100%. The pooled test should therefore be limited by ordinary sampling noise rather than by a structural cap, and the joint power should cross 80% within the swept range, plausibly between about 70 and 130 participants per language. Second, the limiting quantity changes. It is no longer the verb count but the variation between languages, especially in the interaction, whose pilot values span −0.44 to −0.15 per standard deviation. That variation does not shrink with more verbs or more participants, and with only three languages its variance is only weakly pinned down by the data and rests partly on the prior (Gelman, 2006), so it is the number to watch in the results and in sensitivity checks. Under the pilot values, it still leaves the pooled ceilings near 97% or higher. Third, the interpretation changes. A strong pooled Bayes factor alongside a wide Turkish-specific interval is a coherent and likely outcome, and it is the same pattern the cross-linguistic literature shows. The per-language conclusions above stand unchanged. Norwegian is confirmable on its own, and the English and Turkish per-language joint tests remain capped whatever the pooled result.

If the simulations return materially weaker power than these expectations, the likely cause is the between-language variance component rather than the focal effects, and the pooled test's sensitivity to that component's prior will be examined before any recommendation is drawn from it.

Limitations and Status

The status differs across the two analyses. The literature-anchored cross-check is complete for English and Turkish and points to a modest sample at the full verb set. The data-grounded analysis, which is the primary basis, is substantially complete. Norwegian is finished and is adequately powered at about 100 participants. English and Turkish fall short of the joint power target at every sample size run so far, and only their largest cells are still computing. The finding that participants alone cannot bring English and Turkish to target rests on the affectedness effect being verb-limited. The 400-participant cells now confirm that for English at 32 replicates, and the analytic ceilings bound it from above, so the running cells are expected to confirm the plateau rather than change its direction.

Two aspects of the simulation record qualify how far these figures can be pressed. Each assurance replicate reuses one of the first 24 to 40 saved posterior draws rather than draws spread evenly across the whole posterior. Checked against the full posterior, those blocks are representative for English and Norwegian and sit slightly below the mean for the Turkish affectedness slope, so the simulated Turkish figures are, if anything, mildly pessimistic. The analytic ceilings use all 8,000 draws and are unaffected, and the pooled cross-linguistic design analysis and the decision arm sample their draw indices randomly over the full pilot posteriors, so the limitation does not reach them. Replication itself is complete, since the original single-language grid finished all 720 of its cells and the low counts noticed mid-run were cells still computing.

Each data-grounded condition rests on 22 to 144 simulated studies, and each literature-anchored cell on 50, so the figures carry a Monte-Carlo error of several percentage points near a power of 80%. Individual cells close to the threshold should be read as near target rather than pinned, and the conclusions rest on the pattern across sample sizes rather than on any single cell. On the literature effects, the adequate sample at 72 verbs is reported as at most 50, since the main sweep does not go below 50. It is not pinned down further, as the recommendation rests on the data-grounded analysis rather than on this cross-check.

This is an a-priori power analysis, a simulation under assumed effects to size the study, and it is distinct from the confirmatory analysis of the real ratings that will be run once data collection is complete. The analysis code and configuration are openly available, and the report rebuilds from the committed summary tables without the raw data, which remain private to CLAPS until the project reaches a more advanced stage. The report will be finalised once the data-grounded runs complete and the largest English and Turkish samples are in.

References

Albers, C., & Lakens, D. (2018). When power analyses based on pilot data are biased: Inaccurate effect size estimators and follow-up bias. *Journal of Experimental Social Psychology*, *74*, 187–195. https://doi.org/10.1016/j.jesp.2017.09.004

Ambridge, B., Arnon, I., & Bekman, D. (2023). He was run-over by a bus: Passive, but not pseudo-passive, sentences are rated as more acceptable when the subject is highly affected: New data from Hebrew, and a meta-analytic synthesis across English, Balinese, Hebrew, Indonesian and Mandarin. *Glossa Psycholinguistics*, *2*(1). https://doi.org/10.5070/G6011177

Ambridge, B., Bidgood, A., Pine, J. M., Rowland, C. F., & Freudenthal, D. (2016). Is passive syntax semantically constrained? Evidence from adult grammaticality judgment and comprehension studies. *Cognitive Science*, *40*(6), 1435–1459. https://doi.org/10.1111/cogs.12277

Aryawibawa, I. N., & Ambridge, B. (2018). Is syntax semantically constrained? Evidence from a grammaticality judgment study of Indonesian. *Cognitive Science*, *42*(8), 3135–3148. https://doi.org/10.1111/cogs.12697

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Beavers, J. (2011). On affectedness. *Natural Language & Linguistic Theory*, *29*(2), 335–370. https://doi.org/10.1007/s11049-011-9124-6

Bidgood, A., Pine, J. M., Rowland, C. F., & Ambridge, B. (2020). Syntactic representations are both abstract and semantically constrained: Evidence from children's and adults' comprehension and production/priming of the English passive. *Cognitive Science*, *44*(9), e12892. https://doi.org/10.1111/cogs.12892

Brysbaert, M., & Stevens, M. (2018). Power analysis and effect size in mixed effects models: A tutorial. *Journal of Cognition*, *1*(1), 9. https://doi.org/10.5334/joc.10

Bürkner, P.-C., & Vuorre, M. (2019). Ordinal regression models in psychology: A tutorial. *Advances in Methods and Practices in Psychological Science*, *2*(1), 77–101. https://doi.org/10.1177/2515245918823199

Button, K. S., Ioannidis, J. P. A., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S. J., & Munafò, M. R. (2013). Power failure: Why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience*, *14*(5), 365–376. https://doi.org/10.1038/nrn3475

Clark, H. H. (1973). The language-as-fixed-effect fallacy: A critique of language statistics in psychological research. *Journal of Verbal Learning and Verbal Behavior*, *12*(4), 335–359. https://doi.org/10.1016/S0022-5371(73)80014-3

Darmasetiyawan, I. M. S., & Ambridge, B. (2022). Syntactic representations contain semantic information: Evidence from Balinese passives. *Collabra: Psychology*, *8*(1), 33133. https://doi.org/10.1525/collabra.33133

Dowty, D. R. (1991). Thematic proto-roles and argument selection. *Language*, *67*(3), 547–619. https://doi.org/10.1353/lan.1991.0021

Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, *1*(3), 515–534. https://doi.org/10.1214/06-BA117A

Gelman, A., & Carlin, J. (2014). Beyond power calculations: Assessing Type S (sign) and Type M (magnitude) errors. *Perspectives on Psychological Science*, *9*(6), 641–651. https://doi.org/10.1177/1745691614551642

Graham, J. W., Taylor, B. J., Olchowski, A. E., & Cumsille, P. E. (2006). Planned missing data designs in psychological research. *Psychological Methods*, *11*(4), 323–343. https://doi.org/10.1037/1082-989X.11.4.323

Ioannidis, J. P. A. (2008). Why most discovered true associations are inflated. *Epidemiology*, *19*(5), 640–648. https://doi.org/10.1097/EDE.0b013e31818131e7

Judd, C. M., Westfall, J., & Kenny, D. A. (2017). Experiments with more than one random factor: Designs, analytic models, and statistical power. *Annual Review of Psychology*, *68*, 601–625. https://doi.org/10.1146/annurev-psych-122414-033702

Lewandowski, D., Kurowicka, D., & Joe, H. (2009). Generating random correlation matrices based on vines and extended onion method. *Journal of Multivariate Analysis*, *100*(9), 1989–2001. https://doi.org/10.1016/j.jmva.2009.04.008

Liddell, T. M., & Kruschke, J. K. (2018). Analyzing ordinal data with metric models: What could possibly go wrong? *Journal of Experimental Social Psychology*, *79*, 328–348. https://doi.org/10.1016/j.jesp.2018.08.009

Little, T. D., & Rhemtulla, M. (2013). Planned missing data designs for developmental researchers. *Child Development Perspectives*, *7*(4), 199–204. https://doi.org/10.1111/cdep.12043

Liu, L., & Ambridge, B. (2021). Balancing information-structure and semantic constraints on construction choice: Building a computational model of passive and passive-like constructions in Mandarin Chinese. *Cognitive Linguistics*, *32*(3), 349–388. https://doi.org/10.1515/cog-2019-0100

Perugini, M., Gallucci, M., & Costantini, G. (2014). Safeguard power as a protection against imprecise power estimates. *Perspectives on Psychological Science*, *9*(3), 319–332. https://doi.org/10.1177/1745691614528519

Pinker, S., Lebeaux, D. S., & Frost, L. A. (1987). Productivity and constraints in the acquisition of the passive. *Cognition*, *26*(3), 195–267. https://doi.org/10.1016/S0010-0277(87)80001-X

Rouder, J. N. (2014). Optional stopping: No problem for Bayesians. *Psychonomic Bulletin & Review*, *21*(2), 301–308. https://doi.org/10.3758/s13423-014-0595-4

Schönbrodt, F. D., Wagenmakers, E.-J., Zehetleitner, M., & Perugini, M. (2017). Sequential hypothesis testing with Bayes factors: Efficiently testing mean differences. *Psychological Methods*, *22*(2), 322–339. https://doi.org/10.1037/met0000061

Stefan, A. M., Gronau, Q. F., Schönbrodt, F. D., & Wagenmakers, E.-J. (2019). A tutorial on Bayes factor design analysis using an informed prior. *Behavior Research Methods*, *51*(3), 1042–1058. https://doi.org/10.3758/s13428-018-01189-8

Taylor, J. E., Rousselet, G. A., Scheepers, C., & Sereno, S. C. (2023). Rating norms should be calculated from cumulative link mixed effects models. *Behavior Research Methods*, *55*(5), 2175–2196. https://doi.org/10.3758/s13428-022-01814-7

Vasishth, S., Mertzen, D., Jäger, L. A., & Gelman, A. (2018). The statistical significance filter leads to overoptimistic expectations of replicability. *Journal of Memory and Language*, *103*, 151–175. https://doi.org/10.1016/j.jml.2018.07.004

Veríssimo, J. (2021). Analysis of rating scales: A pervasive problem in bilingualism research and a solution with Bayesian ordinal models. *Bilingualism: Language and Cognition*, *24*(5), 842–848. https://doi.org/10.1017/S1366728921000316

Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage-Dickey method. *Cognitive Psychology*, *60*(3), 158–189. https://doi.org/10.1016/j.cogpsych.2009.12.001

Westfall, J., Kenny, D. A., & Judd, C. M. (2014). Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *Journal of Experimental Psychology: General*, *143*(5), 2020–2045. https://doi.org/10.1037/xge0000014

Appendix A

Composition of the Pilot Sample

This appendix describes the pilot sample from which the affectedness ratings and the descriptive trends in Figure 1 are drawn, and against which the data-grounded analysis fits its models. The counts are read directly from the harmonised pilot data.

Table 4

Composition of the Harmonised Pilot Sample, by Language

Language	Participants	Verbs	Sentence types	Items	Judgements
English	70	72	3	144	30,240
Turkish	70	72	3	144	30,240
Norwegian	57	66	2	132	15,048

Note. Counts are read from the harmonised pilot data. Items are verb-by-agent-gender combinations, so the item count is twice the verb count. Norwegian lacks a pseudo-passive, hence two sentence types rather than three.

Appendix B

Key Code for the Simulation and the Models

This appendix reproduces the central code, lightly abridged, from the analysis scripts. The code shown is the data-generating process for the literature-anchored cross-check, in `R/06_simulate_design.R`, together with the analysis model in `R/04_model_formulas.R` and the Bayes factor in `R/05_hypothesis_tests.R`. The data-grounded analysis uses the same analysis model and Bayes factor, but draws its data-generating effects, random-effect covariance and thresholds from the fitted pilot models, and reads each verb's affectedness from the pilot data. Its assurance and safeguard variants are in `R/10_simulate_from_pilot.R` and `R/10_simulate_from_pilot_v2.R`.

Each simulated dataset is drawn from the same cumulative-logit model that is later fitted. In the cross-check, affectedness is drawn once per verb across a plausible range, which makes it a verb-level predictor, and the ordinal response follows from the cumulative-logit linear predictor.

```
# DGP for the literature-anchored cross-check, abridged from
 ↪ R/06_simulate_design.R.
verb_semantics <- runif(n_verbs, min = -0.5, max = 0.5)   # one value per verb

# For participant pid, verb vid and sentence type st:
sem   <- verb_semantics[[vid]]
b_int <- if (st == "Active")                 beta_active_interaction * sem
         else if (st == "Pseudo_Passive") beta_pseudo_interaction * sem
         else 0
eta   <- part_intercepts[pid] + part_slopes[pid] * sem +
         verb_intercepts[vid] + beta_semantics * sem + b_int

cum_probs <- plogis(thresholds - eta)        # seven ordered categories
probs     <- diff(c(0, cum_probs, 1))
resp      <- sample(seq_len(7), size = 1, prob = probs)
```

The analysis model is the maximal random-effects structure consistent with the design, fitted with the cumulative-logit family.

```
# Analysis model, from R/04_model_formulas.R.
brms::bf(
  Response ~ S_Type * Semantics_scaled +
```

```
    (1 + S_Type * Semantics_scaled | Participant) +
    (1 + S_Type | Verb),
  family = brms::cumulative(link = "logit", threshold = "flexible")
)
```

Evidence for each prediction is a directional Savage-Dickey Bayes factor, with the prior sampled alongside the posterior and the posterior mass taken in the predicted direction.

```
# Directional Savage-Dickey Bayes factor, from R/05_hypothesis_tests.R.
# post_vals and prior_vals are draws of the focal coefficient; the negative
# direction is shown, as for the primary prediction H1b.
post_prob  <- mean(post_vals  < 0)
prior_prob <- mean(prior_vals < 0)
bf_10      <- (post_prob / (1 - post_prob)) / (prior_prob / (1 - prior_prob))
```

Appendix C

Full Bayes-Factor Operating Characteristics

This appendix reports the operating characteristics for all four predictions in every available cell of the corrected literature-anchored run with at least ten replicates, namely the probability of exceeding the primary and secondary Bayes-factor thresholds, the median Bayes factor and the convergence rate, set out in Table 5, which covers English, Norwegian and Turkish.

Table 5

Probability of Exceeding the Primary and Secondary Bayes-Factor Thresholds, Median Bayes Factor and Convergence Rate, by Design Condition and Prediction

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
English	40	50	H1a	74%	96%	2.8e+01	100%	50
English	40	50	H1b	76%	88%	2.1e+01	100%	50
English	40	50	H2a	46%	74%	7.7e+00	100%	50
English	40	50	H2b	0%	0%	1.3e-01	100%	50
English	40	60	H1a	94%	100%	5.3e+01	100%	50
English	40	60	H1b	80%	96%	3.7e+01	100%	50
English	40	60	H2a	68%	94%	2.5e+01	100%	50
English	40	60	H2b	0%	0%	4.1e-02	100%	50
English	40	70	H1a	76%	98%	3.1e+01	98%	50
English	40	70	H1b	86%	96%	8.0e+01	98%	50
English	40	70	H2a	60%	86%	1.6e+01	98%	50
English	40	70	H2b	0%	2%	6.3e-02	98%	50
English	40	80	H1a	80%	98%	2.6e+01	98%	50
English	40	80	H1b	96%	98%	1.8e+02	98%	50
English	40	80	H2a	52%	88%	1.1e+01	98%	50
English	40	80	H2b	0%	0%	9.3e-02	98%	50
English	40	90	H1a	88%	100%	4.9e+01	98%	50
English	40	90	H1b	96%	100%	5.1e+02	98%	50
English	40	90	H2a	58%	88%	1.2e+01	98%	50
English	40	90	H2b	0%	0%	8.1e-02	98%	50
English	40	100	H1a	78%	94%	9.7e+01	96%	50
English	40	100	H1b	94%	100%	7.6e+02	96%	50
English	40	100	H2a	42%	82%	8.6e+00	96%	50
English	40	100	H2b	0%	0%	1.2e-01	96%	50
English	40	120	H1a	84%	96%	6.2e+01	88%	50
English	40	120	H1b	98%	98%	3.2e+03	88%	50
English	40	120	H2a	68%	82%	2.0e+01	88%	50
English	40	120	H2b	0%	0%	4.9e-02	88%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
English	72	50	H1a	98%	100%	7.6e+02	92%	50
English	72	50	H1b	98%	100%	8.6e+02	92%	50
English	72	50	H2a	58%	84%	1.5e+01	92%	50
English	72	50	H2b	0%	0%	6.7e-02	92%	50
English	72	60	H1a	100%	100%	1.6e+03	90%	50
English	72	60	H1b	98%	100%	1.2e+03	90%	50
English	72	60	H2a	72%	92%	1.9e+01	90%	50
English	72	60	H2b	0%	2%	5.2e-02	90%	50
English	72	70	H1a	98%	100%	5.8e+02	84%	50
English	72	70	H1b	98%	100%	5.9e+03	84%	50
English	72	70	H2a	60%	86%	1.5e+01	84%	50
English	72	70	H2b	0%	0%	6.6e-02	84%	50
English	72	80	H1a	96%	100%	7.1e+02	82%	50
English	72	80	H1b	100%	100%	7.8e+03	82%	50
English	72	80	H2a	78%	94%	5.5e+01	82%	50
English	72	80	H2b	0%	2%	1.8e-02	82%	50
English	72	90	H1a	100%	100%	1.8e+03	82%	50
English	72	90	H1b	100%	100%	9.8e+05	82%	50
English	72	90	H2a	80%	88%	5.8e+01	82%	50
English	72	90	H2b	0%	0%	1.7e-02	82%	50
English	72	100	H1a	96%	100%	9.4e+02	92%	50
English	72	100	H1b	100%	100%	9.7e+05	92%	50
English	72	100	H2a	78%	90%	5.1e+01	92%	50
English	72	100	H2b	0%	0%	1.9e-02	92%	50
English	72	120	H1a	100%	100%	4.8e+02	90%	50
English	72	120	H1b	100%	100%	9.9e+05	90%	50
English	72	120	H2a	86%	94%	5.6e+01	90%	50
English	72	120	H2b	0%	0%	1.8e-02	90%	50
Norwegian	40	50	H1a	78%	94%	2.6e+01	100%	50
Norwegian	40	50	H1b	62%	86%	1.9e+01	100%	50
Norwegian	40	60	H1a	68%	90%	3.8e+01	96%	50
Norwegian	40	60	H1b	90%	96%	6.4e+01	96%	50
Norwegian	40	70	H1a	72%	94%	2.7e+01	100%	50
Norwegian	40	70	H1b	88%	98%	1.1e+02	100%	50
Norwegian	40	80	H1a	76%	90%	4.0e+01	96%	50
Norwegian	40	80	H1b	96%	100%	2.1e+02	96%	50
Norwegian	40	90	H1a	82%	94%	3.4e+01	98%	50
Norwegian	40	90	H1b	96%	100%	2.8e+02	98%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
Norwegian	40	100	H1a	86%	96%	4.9e+01	98%	50
Norwegian	40	100	H1b	94%	100%	8.5e+02	98%	50
Norwegian	40	120	H1a	86%	100%	3.5e+01	98%	50
Norwegian	40	120	H1b	100%	100%	1.4e+03	98%	50
Norwegian	72	50	H1a	92%	98%	5.5e+02	98%	50
Norwegian	72	50	H1b	94%	98%	3.4e+02	98%	50
Norwegian	72	60	H1a	96%	100%	4.4e+02	96%	50
Norwegian	72	60	H1b	100%	100%	5.7e+02	96%	50
Norwegian	72	70	H1a	100%	100%	7.0e+02	96%	50
Norwegian	72	70	H1b	100%	100%	2.0e+03	96%	50
Norwegian	72	80	H1a	100%	100%	4.1e+02	94%	50
Norwegian	72	80	H1b	100%	100%	8.0e+03	94%	50
Norwegian	72	90	H1a	98%	100%	1.1e+03	84%	50
Norwegian	72	90	H1b	100%	100%	4.9e+05	84%	50
Norwegian	72	100	H1a	96%	100%	5.8e+02	88%	50
Norwegian	72	100	H1b	100%	100%	9.9e+05	88%	50
Norwegian	72	120	H1a	98%	100%	1.1e+03	90%	50
Norwegian	72	120	H1b	100%	100%	9.9e+05	90%	50
Turkish	40	50	H1a	74%	95%	6.2e+01	100%	19
Turkish	40	50	H1b	79%	95%	1.3e+02	100%	19
Turkish	40	50	H2a	26%	74%	6.4e+00	100%	19
Turkish	40	50	H2b	0%	0%	1.6e-01	100%	19
Turkish	40	60	H1a	74%	90%	7.3e+01	100%	50
Turkish	40	60	H1b	86%	96%	7.9e+01	100%	50
Turkish	40	60	H2a	54%	74%	1.1e+01	100%	50
Turkish	40	60	H2b	2%	2%	8.9e-02	100%	50
Turkish	40	70	H1a	88%	98%	5.4e+01	98%	50
Turkish	40	70	H1b	90%	98%	5.9e+01	98%	50
Turkish	40	70	H2a	58%	86%	1.3e+01	98%	50
Turkish	40	70	H2b	0%	0%	7.5e-02	98%	50
Turkish	40	80	H1a	80%	96%	4.3e+01	98%	50
Turkish	40	80	H1b	94%	98%	1.8e+02	98%	50
Turkish	40	80	H2a	70%	94%	3.1e+01	98%	50
Turkish	40	80	H2b	0%	0%	3.2e-02	98%	50
Turkish	40	90	H1a	90%	94%	6.7e+01	86%	50
Turkish	40	90	H1b	98%	98%	4.0e+02	86%	50
Turkish	40	90	H2a	62%	82%	2.1e+01	86%	50
Turkish	40	90	H2b	0%	0%	5.2e-02	86%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
Turkish	40	100	H1a	88%	96%	4.4e+01	100%	50
Turkish	40	100	H1b	98%	100%	9.8e+02	100%	50
Turkish	40	100	H2a	70%	86%	3.0e+01	100%	50
Turkish	40	100	H2b	0%	2%	3.3e-02	100%	50
Turkish	72	50	H1a	96%	98%	1.8e+03	90%	50
Turkish	72	50	H1b	96%	100%	8.8e+02	90%	50
Turkish	72	50	H2a	58%	86%	1.6e+01	90%	50
Turkish	72	50	H2b	0%	0%	6.4e-02	90%	50
Turkish	72	60	H1a	100%	100%	4.8e+02	94%	50
Turkish	72	60	H1b	96%	100%	1.0e+03	94%	50
Turkish	72	60	H2a	62%	92%	2.1e+01	94%	50
Turkish	72	60	H2b	0%	0%	4.9e-02	94%	50
Turkish	72	70	H1a	96%	100%	4.1e+02	88%	50
Turkish	72	70	H1b	100%	100%	8.0e+03	88%	50
Turkish	72	70	H2a	70%	90%	2.6e+01	88%	50
Turkish	72	70	H2b	0%	0%	3.8e-02	88%	50
Turkish	72	80	H1a	100%	100%	1.3e+03	98%	50
Turkish	72	80	H1b	100%	100%	9.7e+05	98%	50
Turkish	72	80	H2a	68%	86%	2.4e+01	98%	50
Turkish	72	80	H2b	0%	0%	4.1e-02	98%	50
Turkish	72	90	H1a	96%	100%	7.5e+02	94%	50
Turkish	72	90	H1b	100%	100%	9.7e+05	94%	50
Turkish	72	90	H2a	68%	88%	3.3e+01	94%	50
Turkish	72	90	H2b	0%	0%	3.1e-02	94%	50
Turkish	72	100	H1a	96%	96%	1.4e+03	82%	50
Turkish	72	100	H1b	100%	100%	9.9e+05	82%	50
Turkish	72	100	H2a	88%	96%	4.9e+01	82%	50
Turkish	72	100	H2b	0%	0%	2.0e-02	82%	50
Turkish	72	120	H1a	100%	100%	5.5e+02	88%	50
Turkish	72	120	H1b	100%	100%	9.9e+05	88%	50
Turkish	72	120	H2a	84%	100%	7.8e+01	88%	50
Turkish	72	120	H2b	0%	0%	1.3e-02	88%	50

Note. The probability columns give the proportion of simulated studies reaching a Bayes factor of 10 and of 3, Median BF the median across them to two significant figures, Convergence the proportion of fits meeting the criteria, and Reps the number of studies in the cell. Cells with fewer than ten are omitted as still accumulating, and Norwegian has no pseudo-passive, hence half as many rows.

Appendix D

Guide to the Analysis Code

This appendix maps the analysis code, which is openly available at https://github.com/pablobernabeu/CLAPS-analysis. It is meant to make the repository navigable for a reader who wants to check how a number in this report was produced, or to reuse part of the machinery. The raw pilot data are not published, since they are participant-level records held privately until the project is further advanced, so the repository ships the simulation and analysis code together with the aggregated summaries the report reads.

Three entry points cover most purposes. A reader checking a reported number wants `outputs/`, which holds the aggregated summaries and is where every figure in this report comes from. A reader checking how a simulated study is built and analysed wants `R/06_simulate_design.R` for the literature-anchored generator, `R/10_simulate_from_pilot_v2.R` for the pilot-grounded one, and `R/05_hypothesis_tests.R` for the Bayes factor. A reader intending to rerun the analysis wants `scripts/`, whose files are numbered in the order they are meant to run, together with the matching submitters in `hpc/`. Table 6 sets out what each top-level directory holds.

Table 6

Layout of the public analysis repository.

Location	What it holds
`R/`	The building blocks: data validation and preprocessing (`01`, `02`), priors and model formulas (`03`, `04`), the Bayes factor (`05`), the two simulation engines (`06`, and `10` with `11`), convergence diagnostics (`07`) and summarising (`08`)
`scripts/`	Runnable steps, numbered in running order, from harmonising the pilot (`00`) through fitting the pilot models, generating a design grid, running one cell, to aggregating results (`aggregate_pilot.R`, `aggregate_pooled.R`, `aggregate_literature.R`)
`hpc/`	One SLURM submitter per arm. Each takes a grid file and an output directory, and runs one grid row per array task
`config/`	The design grids, one row per simulated condition, plus the model and prior configuration
`outputs/`	The aggregated summaries this report reads, and the figure it displays
`tests/`	Unit tests for the validation, prior, diagnostic and Bayes-factor logic
`reports/`	This report, its bibliography and citation style

The unit of work is the cell, meaning one simulated condition: a language, a participant count, a verb count and a replicate index. A grid file in `config/` lists the cells, an array task runs one row, and each cell is saved as a single `.rds` file holding its summary, its Bayes factors and its convergence diagnostics. The aggregation scripts read a directory of those files and reduce them to the tables in `outputs/`. Reading one saved cell is therefore the quickest way to see exactly what the pipeline computes per simulated study.

Two conventions are worth knowing before reading the code. Names carrying `pilot` refer to the data-grounded analysis that conditions on the fitted pilot models and forms the primary basis of this report, while names carrying `literature` refer to the cross-check that conditions on the published effect sizes. Separately, the suffix `_v2` on a simulation engine marks the extended version rather than a correction, so both files are live and the pair should be read together.

Three files in `R/` generate simulated studies, `06`, `10` and `11`, and between them they implement two engines rather than three. The engines differ in where the true effects come from, and the second of them is written at two scopes.

The literature-anchored engine is `R/06_simulate_design.R`. It invents a study from assumed effect sizes taken from the published estimates, drawing participants, verbs and affectedness values from scratch. Because the parameters are stipulated rather than estimated, extending it from one language to several is only a matter of adding a language level to the same recipe, which is why this engine covers both scopes in one file: `simulate_claps_data()` for a single language and `simulate_claps_data_multilanguage()` beside it.

The pilot-grounded engine is `R/10_simulate_from_pilot.R`. It takes the true effects from a model already fitted to the pilot data, so a simulated study inherits that language's estimated slopes, thresholds and variance components together with its real verbs and their measured affectedness. `R/10_simulate_from_pilot_v2.R` loads that file and adds the assurance and safeguard variants, which draw the effects from the pilot posterior rather than fixing them at its mean.

Extending this second engine across languages is not a matter of adding a level, because each language carries its own fitted model. A pooled study has to be assembled from three separate sets of parameters, simulated language by language and then combined before a single cross-language model is fitted to the result. That orchestration is what `R/11_simulate_pooled_v2.R` adds. It loads the `_v2` file, calls its per-language simulator once for each language, binds the three datasets and fits the pooled model, so it is the same engine at a wider scope rather than a third method.

The files therefore stack rather than compete, each loading the one below it and adding to it: `11` on `10_v2` on `10`, and separately `R/06_simulate_design_gelman.R` on `06`. That last file is the narrowest of them. It changes only the scale on which the analysed predictor is expressed, leaving the simulated responses identical, so that the literature-anchored cells are analysed on the same footing as the confirmatory model.

Both engines end in the same place. Each defines a cell runner, `run_design_cell()` in `06` and `run_databased_cell_v2()` and `run_pooled_cell_v2()` in `10_v2` and `11`, which simulates one dataset, fits it with `R/04_model_formulas.R` and the priors in `R/03_define_priors.R`, computes the Bayes factors with `R/05_hypothesis_tests.R` and records the diagnostics from `R/07_extract_diagnostics.R`, then writes the `.rds` for that cell. Only the source of the true effects differs; the model, the evidence criterion and the convergence standard are shared, which is what makes the two analyses comparable. Table 7 summarises where each engine's effects come from and which file implements it at each scope.

Table 7

The two simulation engines and the files implementing them.

Engine	Where the effects come from	Single language	All three pooled
Literature-anchored	Published effect sizes, stipulated	`06`	`06`, the same file
Pilot-grounded	Estimated from the fitted pilot models	`10`, extended by `10_v2`	`11`, which loads `10_v2`

Following one arm end to end is the quickest way to see how the pieces connect. For the pooled analysis reported here, the grid is `config/design_grid_pooled_v2.csv`, one row per simulated study; `hpc/submit_pooled_v2_array.sh` submits one array task per row; `scripts/07_pooled_cell_v2.R` runs a single row by calling `run_pooled_cell_v2()`; and `scripts/aggregate_pooled.R` reduces the resulting `.rds` files to `pooled_power_pilot.csv` in `outputs/design_summary_pilot/`, which is the table this report reads for every pooled figure it quotes.